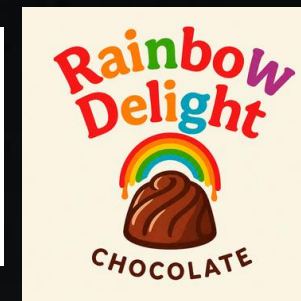
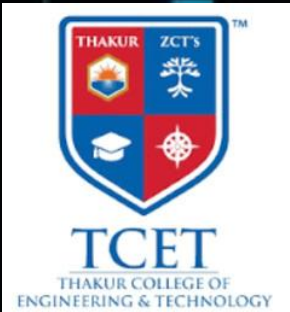




# THINK AI 4.0

*PS 10: AI-Powered Accessibility Assistant*

*Team Name: Team Vaachak*

# *Meet The Team*

**A**

**Anvesh Chafle**

Backend, AI Pipeline &  
Safety Rule Engine

**M**

**Maulik Parshionikar**

Frontend, UI/UX &  
Accessibility

**N**

**Nimisha Jain**

Presenter · Research  
& Live Demo

**J**

**Jiya Khut**

Presenter · Pitch  
& User Testing

# PROBLEM STATEMENT

Prakash-kaka can hear the bill.  
Nobody tells him what to do.

68, cataract, reads only Marathi. The MSEDCL bill, a new medicine strip and bank SMSes arrive in English jargon and small print while his children are at work. He guesses, misses due dates, takes expired medicine, or trusts a fake 'power will be cut tonight' SMS. Google Lens reads the text aloud, but only if he knows what to ask.

Elders need a decision they can trust, not text read aloud.

149 M

Indians aged 60+ today, 347 M by 2050  
UNFPA India Ageing Report 2023

68%

of women aged 60–75 cannot read  
UNFPA India Ageing Report 2023

13.8%

of Indians aged 50+ have visual  
impairment  
National Blindness & VI Survey 2015–19

₹22,495 Cr

lost to cyber fraud in 2025; fake bill SMS  
target elders  
I4C data, via Moneylife

PROPOSED SOLUTION

# VAACHAK वाचक

A voice-first reader that tells elders what to do, and keeps them safe.

Instead of reading text aloud and waiting for the right question, Vaachak turns any bill, medicine strip, notice or message into one spoken action card: what it is, what to do, by when, how much, and the danger.

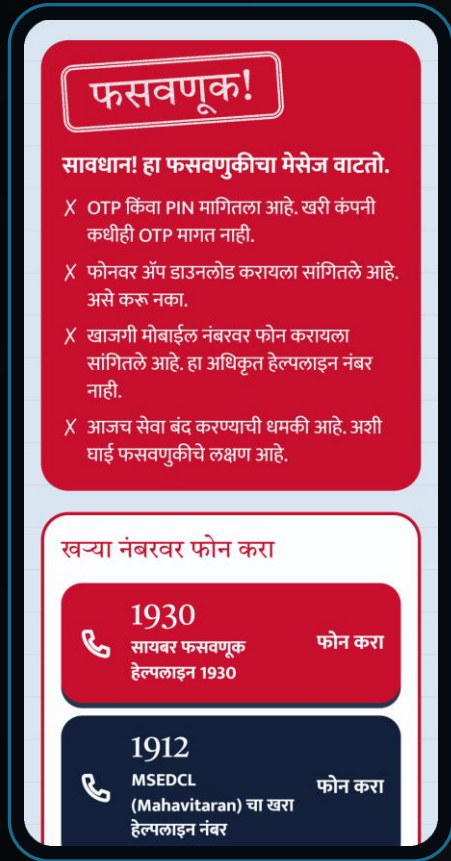


KEY IDEA

Vaachak turns a document from text read aloud into a decision: **what to do, by when, and what is dangerous.**

BUILT & RUNNING TODAY

# LIVE PROTOTYPE



**PROTECT**  
**Fake power-cut SMS**  
Red scam card, one tap to 1930



**VERIFY**  
**MSEDCL bill**  
₹840 by 10 Oct, found in the print



**REMIND**  
**3 medicines**  
One schedule each, into the calendar



**REMEMBER**  
**My papers**  
What is due, scams caught

MEASURED ON THE LIVE APP · 8 OCT 2026

**≈ 2 s**  
pasted SMS → spoken card

**≈ 4 s**  
bill photo → verified card

**≈ 1.5 s**  
until the voice starts

**4 / 4**  
test scams caught · 0 real SMS flagged

**58**  
automated tests on the rules

**3**  
languages: मराठी · हिंदी · English



**Scan & try it**  
[vaachak-zeta.vercel.app](https://vaachak-zeta.vercel.app)  
Tap a sample: works even with Wi-Fi off.

# Tech Stack

## AI / MODEL LAYER

Reads documents, answers questions, speaks.



Gemini 3.8



Sarvam voice



7-model fallback



Voice Q&A

## SAFETY LAYER · NO AI

Checks every ₹, date and scam. Tested.



Rule engine



Tesseract OCR



Scam shield



50 medicines

## APP LAYER

Elder-first PWA: giant text, voice, offline.



React



Vite



Tailwind



PWA

## PLATFORM & ACTIONS

Serverless in Mumbai; acts in one tap.



Node.js



Vercel



WhatsApp



Calendar

# System Diagram



## USP



### Acts, not just reads

Pay, remind, call 1930 or ask family in one tap.



### Scam shield for India

Fake power-cut SMS, KYC, OTP, AnyDesk, personal UPI.



### Works offline

On-phone OCR, 50 medicines and the same safety rules.



### Never guesses

₹ and date checked against the print; unsure means "ask someone".



### Looks after you

Whole-course dose reminders, due dates in My papers, cheaper generic.



### Your language & voice

Marathi, Hindi, English; spoken tour; giant UI.

## Future Scope

Phase  
**1**

### Live today

Bills, medicines, scams and reminders in Marathi, Hindi, English.

Phase  
**2**

### 22 languages + WhatsApp bot

Forward any document to Vaachak; Sarvam / Bhashini voices.

Phase  
**3**

### Family & payments

Caregiver alerts for due bills and scams; BBPS verified payments.

**Future**

### Fully offline Vaachak

On-device AI model; government-scheme notices as checklists.



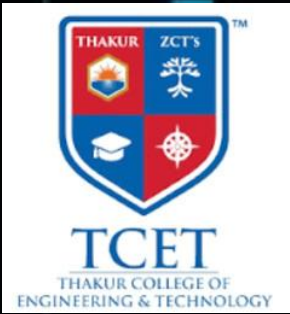
## Prototype Link

live, tested, open on any phone

[vaachak-zeta.vercel.app](https://vaachak-zeta.vercel.app)

Tap a sample (works offline) · tap ? for a spoken tour · paste a scam SMS





# Thank You

Gemini answers questions. **Vaachak** never needs one.

Team Vaachak · [vaachak-zeta.vercel.app](https://vaachak-zeta.vercel.app)